

How a Bank's Capital Requirement Evolves from Basel I to FRTB

A Worked Example Covering Credit, Market, Operational, Counterparty, Leverage and Liquidity Risk

1. The Hypothetical Bank: Meridian Bank

We track one bank, Meridian Bank, through every major regulatory standard. The balance sheet never changes. Only the rules for turning that balance sheet into a capital requirement change. This lets us see exactly what each new standard added. All figures are in billions of dollars.

Balance sheet (\$ billions):

Assets	Amount	Liabilities and Equity	Amount
Cash and OECD government bonds	100	Customer deposits	450
Interbank claims	50	Wholesale and interbank funding	130
Residential mortgages	200	Debt securities issued	50
Corporate loans	250	Capital (equity and subordinated)	50
Trading book (equities, bonds, FX)	80		
Total assets	680	Total liabilities and equity	680

The first four asset lines (600 in total) are the banking book (loans and bonds held to maturity). The trading book (80) is held for short-term trading and carries market risk. The 50 of capital is what the bank actually holds against all of its risks; we keep it fixed and watch whether it stays sufficient as the rules change.

Banking-book risk weights used for the Basel I credit charge:

Asset	Amount	Basel I risk weight
Cash and OECD government bonds	100	0%
Claims on OECD banks (interbank)	50	20%
Residential mortgages	200	50%
Corporate loans	250	100%
Total banking book	600	

Other inputs used later:

- Derivatives book: used with clients for hedging. Small counterparty exposure that becomes a separate charge (CVA) only under Basel III.
- Gross annual income: 20. This is the base for the operational-risk charge under Basel II.
- Total leverage exposure (on balance sheet plus off balance sheet and derivative add-ons): 750.
- Liquidity: high-quality liquid assets 90; stressed 30-day net cash outflows 75; available stable funding 480; required stable funding 450.

Off-Balance-Sheet Items: Exposures That Are Not on the Balance Sheet

Not every exposure sits on the balance sheet. Off-balance-sheet (OBS) items are promises and contracts that create future credit exposure without being recorded as an asset today, because no money has been lent yet. They matter because the promise can turn into a real, funded loan exactly when the borrower is in trouble. Meridian has four common types:

- Loan commitments and undrawn credit lines: the bank has promised to lend up to a limit, and the customer can draw on it later.
- Guarantees and standby letters of credit: the bank promises to pay a third party if its own customer fails to.
- Trade letters of credit: short-term, self-liquidating financing for goods in transit.
- Derivatives (swaps, forwards, options): no principal changes hands, but the bank is exposed if the counterparty defaults on a contract that has moved in the bank's favor.

Regulators do not ignore these. Each item is converted to a credit-equivalent amount using a credit conversion factor (CCF), and that credit-equivalent is then risk-weighted exactly like an on-balance-sheet asset. A higher CCF means the item is more likely to become a real exposure.

Off-balance-sheet item	Notional	Conversion factor	Credit equivalent
Undrawn loan commitments (> 1 year)	100	50%	50
Standby letters of credit (guarantees)	10	100%	10
Trade letters of credit (short-term)	20	20%	4
Interest-rate swaps	200	market value plus add-on	6
Total	330		70

The key difference from on-balance-sheet assets is what is at risk. For a loan you could lose the whole amount, so the exposure is the amount lent. For a swap no principal is at risk: the bank can only lose the cost of replacing the contract if the counterparty defaults, so the credit-equivalent is

the current positive market value plus a small add-on for how much the value could grow before the trade matures.

Why this matters: because OBS items are not recorded as assets, a bank can build up large hidden leverage that the balance sheet does not show. In this example Meridian's on-balance-sheet assets are 680, but once the off-balance-sheet items (about 70 of credit-equivalent) are included its total leverage exposure is 750. This is exactly why Basel III added the leverage ratio, a simple backstop that captures off-balance-sheet exposures the risk-weighted ratios can understate.

Two mechanics used throughout. (1) Risk-weighted assets (RWA) are the common currency: each risk type is converted to an RWA figure, and the minimum capital is a percentage of total RWA. (2) A capital charge in dollars is turned into RWA by multiplying by 12.5 (because $1 / 8\% = 12.5$), so that an 8% ratio on that RWA gives back the original charge.

2. Basel I (1988): Credit Risk Only

Basel I set a single rule: hold at least 8% capital against credit-risk-weighted assets (the Cooke ratio). It ignored market, operational and liquidity risk entirely.

Credit RWA = $100(0\%) + 50(20\%) + 200(50\%) + 250(100\%) = 0 + 10 + 100 + 250 = 360$.

Minimum capital = 8% of 360 = 28.8 (of which Tier 1 at least 4% = 14.4).

Meridian holds 50, so it looks very strong: a surplus of 21.2. But the rule is blind to its trading book, its operations, and its funding. That blindness is what every later standard fixes.

3. 1996 Market Risk Amendment: Adding Market Risk

After trading losses in the early 1990s, regulators added a charge for the trading book. Banks could use an internal model: capital equals a multiplier (at least 3) times the 10-day 99% Value-at-Risk (VaR) of the trading positions.

Assume the trading book has a 10-day 99% VaR of 2.0. Market charge = $3 \times 2.0 = 6.0$. Market RWA = $12.5 \times 6.0 = 75$.

Total RWA = 360 (credit) + 75 (market) = 435. Minimum capital = 8% of 435 = 34.8.

Nothing on the balance sheet changed, yet the requirement rose from 28.8 to 34.8. The trading book now costs capital.

4. Basel II (2004): Risk-Sensitive Credit and a New Operational Charge

Basel II made credit capital sensitive to borrower quality and added a brand-new operational-risk charge (for fraud, systems failures, legal losses and the like). It also introduced the three pillars: minimum capital (Pillar 1), supervisory review (Pillar 2), and market disclosure (Pillar 3).

Credit RWA is recomputed with risk-sensitive weights. Mortgages now get 35% instead of 50%. Corporate loans are split by rating: 150 to A-rated borrowers at 50%, and 100 to unrated or weaker borrowers at 100%.

$$\text{Credit RWA} = 0 + 50(20\%) + 200(35\%) + 150(50\%) + 100(100\%) = 0 + 10 + 70 + 75 + 100 = 255.$$

$$\text{Operational risk (Basic Indicator Approach)} = 15\% \text{ of gross income} = 15\% \times 20 = 3.0.$$

$$\text{Operational RWA} = 12.5 \times 3.0 = 37.5.$$

$$\text{Total RWA} = 255 \text{ (credit)} + 75 \text{ (market)} + 37.5 \text{ (operational)} = 367.5. \text{ Minimum capital} = 8\% \text{ of } 367.5 = 29.4.$$

Notice the trade-off. Better-quality credit lowered credit RWA from 360 to 255, but the new operational charge added 37.5. The total requirement barely moved, but the risk coverage is far wider than under Basel I.

5. Basel 2.5 (2009): Stressed VaR After the Crisis

The 2008 crisis showed that VaR measured in calm markets badly understated trading losses. Basel 2.5 required a second market charge, a Stressed VaR calibrated to a severe past period, added on top of the ordinary VaR charge. In practice this roughly doubled the market-risk capital.

$$\text{New market charge} = \text{ordinary VaR charge} + \text{Stressed VaR charge} = \text{about } 6.0 + 6.0 = 12.0.$$

$$\text{Market RWA} = 12.5 \times 12.0 = 150.$$

$$\text{Total RWA} = 255 \text{ (credit)} + 150 \text{ (market)} + 37.5 \text{ (operational)} = 442.5. \text{ Minimum capital} = 8\% \text{ of } 442.5 = 35.4.$$

The balance sheet still has not changed. The extra capital comes purely from measuring trading risk under stressed conditions instead of calm ones.

6. Basel III (2013 onward): Better Capital, Buffers, Counterparty, Leverage and Liquidity

Basel III kept the same four risk RWAs but changed capital in several ways at once.

- Higher quality: at least 4.5% must be Common Equity Tier 1 (CET1), 6% Tier 1, 8% total.
- A Capital Conservation Buffer of 2.5% (in CET1) sits on top, so the effective total requirement is 10.5%. A countercyclical buffer (0 to 2.5%) and a surcharge for global systemically important banks (G-SIBs) can add more; we assume zero here for simplicity.
- A new CVA charge covers the risk that a derivatives counterparty's credit quality worsens. Assume a CVA charge of 2.0, so CVA RWA = $12.5 \times 2.0 = 25$.

$$\text{Total RWA} = 255 \text{ (credit)} + 150 \text{ (market)} + 37.5 \text{ (operational)} + 25 \text{ (CVA)} = 467.5.$$

Minimum capital including the conservation buffer = 10.5% of 467.5 = 49.1 (of which CET1 at 7.0% = 32.7).

Basel III also added two limits that capital ratios alone never captured:

- Leverage ratio: Tier 1 capital divided by total exposure must be at least 3%. Here that needs $3\% \times 750 = 22.5$ of Tier 1, which is below the risk-based requirement, so the risk-based rule still binds. The leverage ratio is a simple backstop against gaming the risk weights.
- Liquidity Coverage Ratio (LCR): liquid assets divided by 30-day stressed outflows must be at least 100%. Meridian: $90 / 75 = 120\%$. Pass.
- Net Stable Funding Ratio (NSFR): stable funding available divided by stable funding required must be at least 100%. Meridian: $480 / 450 = 107\%$. Pass.

This is the point where a bank could be well capitalized yet still fail, if it could not fund itself for a month. Northern Rock in 2007 had capital but not liquidity. Basel III closed that gap.

7. FRTB (2023): The Trading Book Rebuilt

The Fundamental Review of the Trading Book replaced VaR with Expected Shortfall at 97.5%, which looks deeper into the tail, and added stricter treatment of risk factors that cannot be modelled. For most banks this raised the market-risk charge again. There is also an output floor: total RWA cannot fall below 72.5% of the figure a bank would get using the standardized approach, which limits how far internal models can cut capital.

New market charge rises to about 18.0. Market RWA = $12.5 \times 18.0 = 225$.

Total RWA = 255 (credit) + 225 (market) + 37.5 (operational) + 25 (CVA) = 542.5.

Minimum capital including the conservation buffer = 10.5% of 542.5 = 57.0.

Meridian holds 50. For the first time, that is not enough: it is short by about 7.0. The same bank that looked over-capitalized under Basel I must now raise capital or cut its trading risk.

8. Capital Also Got Better, Not Just Bigger

The requirement did not only grow in amount. Its quality was tightened too. Under Basel I a bank could meet up to half of its 8% with lower-quality Tier 2 instruments. Basel III forces the core to be Common Equity Tier 1 (CET1), the most loss-absorbing form of capital, and applies stricter deductions (for goodwill and deferred tax assets, for example). So the modern requirement is both larger and made of better capital.

Standard	Total minimum	Core (highest-quality)	What the core could be
Basel I (1988)	8%	Tier 1 at least 4%	Common equity plus reserves; up to half of total could be Tier 2
Basel II (2004)	8%	Tier 1 at least 4%	Same structure as Basel I
Basel III (2013)	8% plus 2.5% buffer	CET1 at least 4.5% (7% with buffer)	Common equity only for the core, with tougher deductions

The requirement also became layered. Basel I was a single flat 8%. Basel III stacks a minimum, a capital conservation buffer, and, for larger or more cyclical banks, a countercyclical buffer and a G-SIB surcharge on top. Each layer has a job: the minimum keeps the bank solvent, the buffers give it room to absorb losses without breaching the minimum, and the surcharge makes the most systemic banks hold extra.

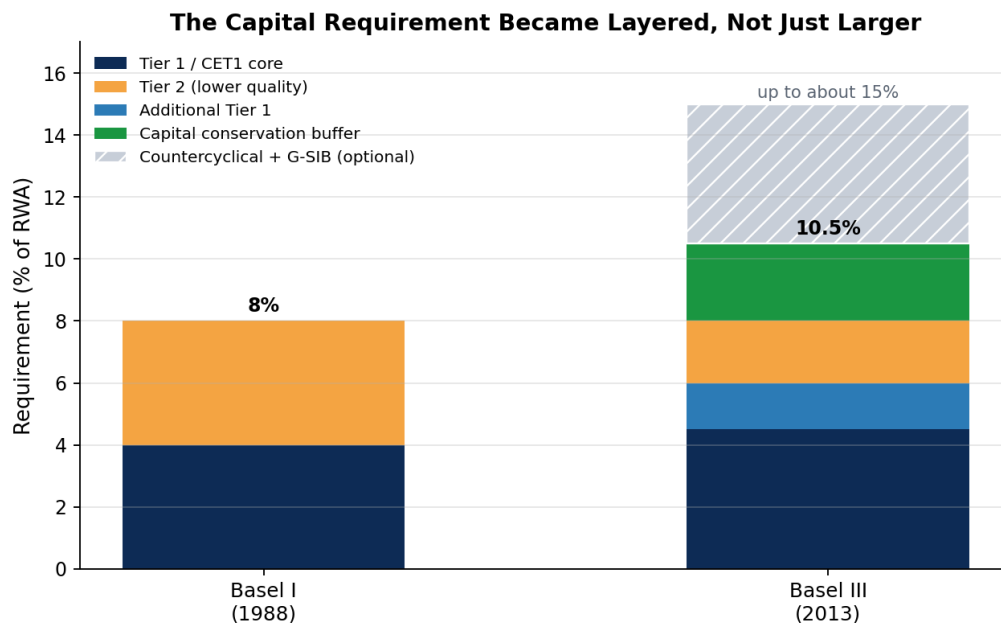


Figure 2. Basel I was a single flat 8% block, and up to half could be lower-quality Tier 2. Basel III is a layered stack built mostly from common equity, plus buffers that can lift the requirement toward 15% for the largest banks.

9. Summary: The Requirement Roughly Doubles

The table shows how each risk type enters over time and how the minimum capital grows. Credit, market, operational and CVA RWA are in billions; the minimum ratio includes the Basel III conservation buffer where it applies.

Standard (year)	Credit	Market	Oper.	CVA	Total RWA	Min ratio	Min capital
Basel I (1988)	360	-	-	-	360	8%	28.8
1996 Amendment	360	75	-	-	435	8%	34.8
Basel II (2004)	255	75	37.5	-	367.5	8%	29.4
Basel 2.5 (2009)	255	150	37.5	-	442.5	8%	35.4
Basel III (2013)	255	150	37.5	25	467.5	10.5%	49.1
FRTB (2023)	255	225	37.5	25	542.5	10.5%	57.0

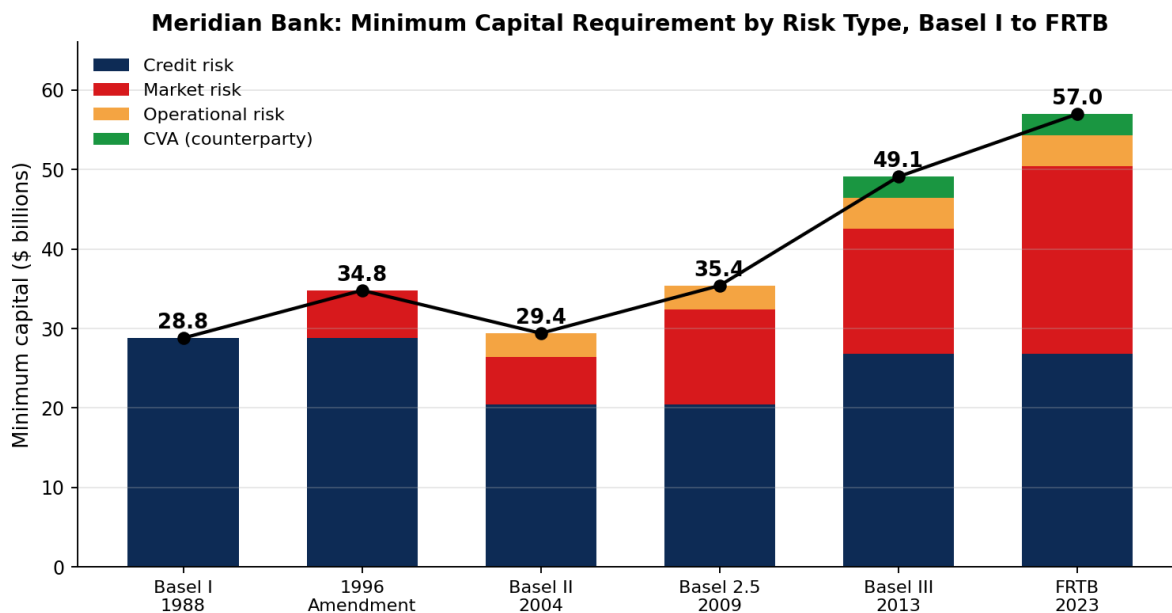


Figure 1. Minimum capital grows from 28.8 to 57.0 as each standard adds a new risk type. The black line is the total; the colored blocks show what drives it.

Reading the same 50 of held capital as a ratio tells the story from the other side. Two columns matter. RWA density (total RWA divided by total assets) rises from 53% to 80%: the same assets are treated as far riskier once every risk type is counted. And the held-capital ratio falls as the identical 50 of capital is spread over more measured risk.

Standard	RWA density (RWA/assets)	Held capital / Total RWA	Status
Basel I (1988)	$360 / 680 = 53\%$	$50 / 360 = 13.9\%$	Strong
1996 Amendment	$435 / 680 = 64\%$	$50 / 435 = 11.5\%$	Comfortable
Basel II (2004)	$367.5 / 680 = 54\%$	$50 / 367.5 = 13.6\%$	Strong
Basel 2.5 (2009)	$442.5 / 680 = 65\%$	$50 / 442.5 = 11.3\%$	Comfortable
Basel III (2013)	$467.5 / 680 = 69\%$	$50 / 467.5 = 10.7\%$	Just above the 10.5% requirement
FRTB (2023)	$542.5 / 680 = 80\%$	$50 / 542.5 = 9.2\%$	Below 10.5%: shortfall of about 7

10. What Each Standard Added, and Why

Each standard was a response to a specific failure. The pattern is simple: a crisis exposes a gap, and the next standard closes it.

Trigger / crisis	Regulatory response
Herstatt Bank collapse (1974); no common capital standard	Basel I (1988)
Rapid growth of derivatives and large trading losses	1996 Market Risk Amendment
Basel I too crude; banks arbitrated the flat risk weights	Basel II (2004)
2007 to 2008 crisis: trading losses far exceeded VaR	Basel 2.5 (2009)
2008 crisis: too little and poor-quality capital, no liquidity rules	Basel III (2013)
Market-risk framework still weak; Basel 2.5 was only a patch	FRTB (2019 to 2023)

In one line each:

- Basel I (1988): credit risk only, a flat 8% of credit RWA. Simple but blind to everything else.
- 1996 Amendment: market risk for the trading book, measured by VaR.
- Basel II (2004): risk-sensitive credit weights, plus a new operational-risk charge, plus the supervisory and disclosure pillars.
- Basel 2.5 (2009): a Stressed VaR add-on, so trading risk is measured under crisis conditions.
- Basel III (2013): higher-quality capital and buffers, a CVA charge for derivative counterparties, a leverage-ratio backstop, and for the first time liquidity rules (LCR and NSFR) and systemic surcharges for the largest banks.
- FRTB (2023): the trading book rebuilt around Expected Shortfall, with an output floor that limits how far internal models can reduce capital.

The single lesson: each standard is a response to the weakness the previous one revealed. The balance sheet of Meridian Bank never moved, yet its minimum capital roughly doubled, from 28.8 to 57.0, as regulators learned to measure risks that Basel I never counted.

Note: all figures are illustrative and rounded for teaching. Real risk weights, model multipliers, buffers and the output floor depend on the bank's jurisdiction, ratings, approvals and portfolio.

11. Glossary

Term	Meaning
RWA	Risk-weighted assets. Each exposure scaled by a regulatory risk weight; the base for the capital ratio.
CET1	Common Equity Tier 1. The highest-quality capital: common shares plus retained earnings.
Tier 1 / Tier 2	Going-concern capital (absorbs losses while the bank operates) and gone-concern capital (absorbs losses in resolution).
CCF	Credit conversion factor. Turns an off-balance-sheet notional into an on-balance-sheet credit-equivalent amount.
VaR	Value-at-Risk. The loss not exceeded at a stated confidence over a set horizon.
ES	Expected Shortfall. The average loss in the tail beyond the VaR point; used by FRTB.
CVA	Credit Valuation Adjustment. Capital for the risk that a derivatives counterparty's credit quality worsens.
LCR	Liquidity Coverage Ratio. Liquid assets divided by 30-day stressed cash outflows; at least 100%.
NSFR	Net Stable Funding Ratio. Stable funding available divided by stable funding required; at least 100%.
G-SIB	Global systemically important bank; carries an extra capital surcharge.